

LeetCode 871: Minimum Number of Refueling Stops

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Problem Understanding

- `stations[i] = [a, b]` where **a** = distance from start position, and **b** = fuel it has.
- The car starts with `startFuel`. The goal is to return the minimum number of refueling stops the car must make to reach the target.
- Assumption: 1 liter of fuel = 1 km of distance.

Intuition & Logic

Should Aadya refill whenever she reaches a gas stop?

No. She should drive until she can't move further, and then decide among all the previous stations which one she should have refilled at.

Strategy: Choose the station with the **max fuel**.

MAX HEAP + GREEDY

- Greedily choose the max available fuel to minimize stops.

Note: Only push reachable stations to your priority queue.

C++ Solution

```
class Solution {
public:
    int minRefuelStops(int target, int startFuel,
vector<vector<int>>& stations) {
        int n=stations.size();
        priority_queue<int> pq;
        int feasible=startFuel;
        int i=0;
        int ans=0;

        while(feasible<target){
            while(i<n && stations[i][0]<=feasible){
                pq.push(stations[i][1]);
                i++;
            }

            if(pq.empty()) return -1;

            feasible+=pq.top();
            pq.pop();
            ans++;
        }
        return ans;
    }
};
```